Design and Implementation of Private Cloud for Higher Education using OpenStack

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Abstract— **Higher Education Institutions requires ready to access dedicated devices and various operating systems to assist in teaching, learning various courses. Due to the high cost of hardware, providing physical devices in ever changing IT field is neither economical nor feasible. A higher education IT course may demand specific software running on the specific operating system at particular hardware configuration. Different operating systems requirement and their complex installation and maintenance process poses difficulty to manage them. OpenStack can address both issues by providing IaaS for hardware resources and standard mechanism to manage and distribute course specific VMs. This paper provides design of private cloud for higher education and proof-of-concept implementation methodology for OpenStack platform. Furthermore, to facilitate the construction of the private cloud, the resource requirement model is formulated by which the requirement of physical resources can be measured for a specific number of virtual machines.**

Keywords— Cloud Computing; OpenStack; IaaS; Open Source; Higher Education;

I. Introduction

Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., servers, storage, networks, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction [1]. Cloud has recently emerged as a powerful system for managing and delivering *Information Technology* (IT) resources on demand for education [2]. Cloud architecture relies on virtualization technology to provide multiple virtual resources on the same physical host. Virtualization is empowering technology that permits a single machine to act as if it were several machines.

The study and use of IT is continuously growing in *Higher Education Institutions* (HEI). IT education requires ready to access dedicated devices. Traditionally, HEI have deployed IT infrastructures in labs and offered IT resources to students using physical devices. Basic IT courses are well supported by dedicated computers but they seem to be limited, as advanced courses demand more than one machine, operating systems, devices and several software tools to study the subjects and perform lab practical. Typically, there is a need to access to various operating systems (Windows, Linux, Kali etc.) and devices (servers, routers, switches, firewall, load-balancer etc.) to assist in learning various IT courses. Due to the high cost of hardware, providing physical devices in ever changing IT field is neither economical nor feasible. Also, the complexity of operating system installation and maintenance poses difficulty to manage them in limited infrastructure. A course may demand specific software running on the specific operating system at particular hardware configuration. One of the potential solutions for this issue is to use hosted hypervisors for virtualization. *Virtual machines* (VM) can be provided by instructors to students based on course requirements. But they have their limitation too i.e., they are limited to host *operating system* (OS) only. The downloading, installing, configuring the OS images and required software is a time consuming process. Beside this, loading and running of VM by students take time and require high configuration systems. Students need to have VM available with them all the time, which is not practical.

A private cloud can provide a solution to both issues by providing virtual hardware resources (servers, routers, networks etc.) and standard mechanism to manage and distribute VM. A private cloud is a type of cloud computing implemented and managed within an organization. A private cloud can deliver *Infrastructure-as-a-Service* (IaaS) to its users. IaaS provides raw computing infrastructure, including storage, processing, and networking resources to consumers. A private cloud is implemented on hardware (e.g. servers, switches) owned by an institution. Private clouds provide pools of virtual resources, sourced from dedicated physical systems that can be automatically provisioned and allocated through a self-service interface. Cloud administrator manages and maintains the underlying cloud infrastructure. A private cloud can be deployed in the organization by a variety of software. Both proprietary and open sources software are available for successful implementation of private cloud.

In this paper, we present the design and implementation of private cloud prototype for higher education using OpenStack. OpenStack is open source software for building public and private clouds. OpenStack provides an IaaS solution through a variety of services. It is a cloud operating system that controls large pools of virtual servers, storage, and networking resources. All services are controlled through a dashboard or API. OpenStack aims at scalability without complexity [3]. We have chosen OpenStack to deploy the private cloud because it is a dominant platform and becoming the standard for private cloud deployment [4].

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The paper is organized as follows. Section 2 gives an overview of OpenStack services and its architecture. Section 3 presents the related work done. Section 4 presents a brief description of our proposed private cloud design. Section 5 presents the implementation methodology and resource requirement model. The conclusion is given in Section 6.

II. OPENSTACK ARCHITECTURE OVERVIEW

OpenStack provides a cloud-based IaaS, which is modular and scalable. We have selected seven core modules of OpenStack [5-6]. Fig. 1, provides a high-level overview of the OpenStack basic services and their relationship with each other.

A. *OpenStack Services(Code Name)*

1) Dashboard (Horizon) is a graphical user interface for users and administrators to perform operations such as creating and launching instances, managing networking, and setting access control. The Dashboard service provides the Project, Admin, and Settings default dashboards. The modular design enables the dashboard to interface with other products such as billing, monitoring, and additional management tools. The identity of the logged-in user determines the dashboards and panels that are visible within Horizon.

2) Identity (Keystone) is a common authentication and authorization system for all OpenStack components. It provides a central list of users. Keystone supports various forms of authentication mechanisms, including user name and password credentials, token-based systems, and AWS-style log-ins. Keystone also provides a central catalog of services and endpoints running in the OpenStack cloud, which acts as a service directory for OpenStack systems.

3) Networking (Neutron) OpenStack networking allows users to create their own networks and connect devices and servers to one or more networks. Neutron creates and manages virtual networking infrastructure. Infrastructure elements include networks, subnets, and routers. Users can create networks, control traffic, and connect servers and devices to one or more networks. Users can also deploy advanced services such as firewalls, load balancer or virtual private networks. IPs can be either dedicated or floating; floating IPs allows dynamic traffic rerouting.

4) Block Storage (Cinder) provides persistent block storage management for virtual hard drives. Block Storage enables the user to create and delete block devices and to manage attachment of block devices to servers. The life cycle of Cinder volumes is maintained independent of compute instances, and volumes may be attached or detached to one or more compute instances to provide a backing store for file system based storage.

5) Compute (Nova) serves as the core of the OpenStack cloud by providing VMs on demand. Its main goal is to manage VM functions like creating, starting, stopping, and so on. All provisioning is image-based and the OpenStack Image Service (Glance) is a prerequisite for the Compute service. Nova interacts with Keystone for authentication, Glance for images, and Horizon for the user and administrative interface. Access to images is limited by project and by the user; quotas are limited per project (for example, the number of instances).

6) Image (Glance) acts as a registry for virtual disk images. Users can add new images or take a snapshot of an existing server for immediate storage. The user can use the snapshots for backup or as templates for new servers. Registered images can be stored in the Swift service, as well as in other locations. Some common image formats supported are raw, iso, qcow2, vhd, vdi, vmdk.

7) Object Storage (Swift) provides an HTTP-accessible storage system for large amounts of data, including static entities such as videos, images, email messages, files, or VM images. Objects are stored as binaries on the underlying file system along with metadata stored in the extended attributes of each file. Generally, object storage provides access through an API. Object storage can distribute requests across a number of storage hosts. All objects are accessible in one single namespace, and object storage systems are usually highly scalable.

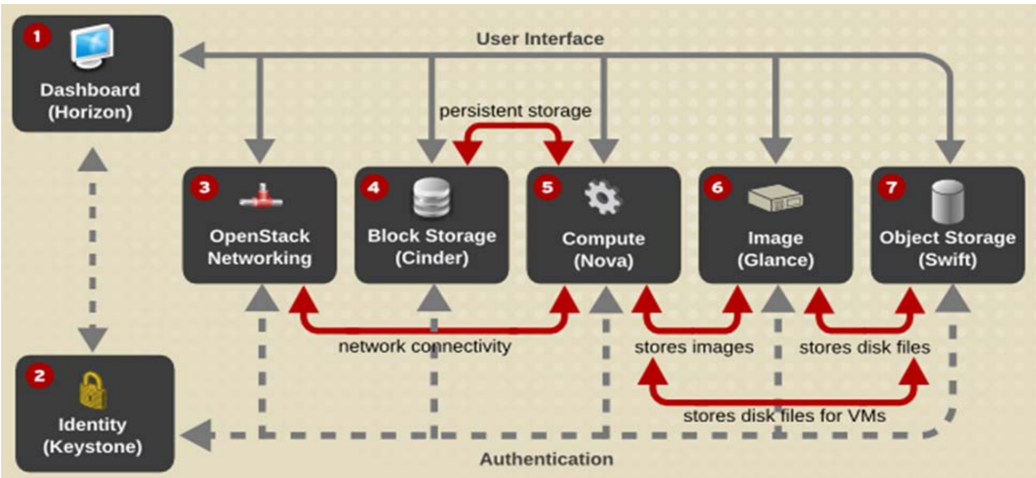

Fig. 1. OpenStack Conceptual Architecture [4].

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III. RELATED WORK

There has been significant work on private cloud based virtual labs using OpenStack [7 - 14]. In [7], authors designs and develop a virtual experiment platform based on OpenStack which enables users to choose experiment resource freely and build corresponding topology. Their platform creates relevant virtual devices and establishes network topology according to user demand automatically. Then, users are able to access the experiment resources and perform tests. Their platform renders teachers and students a perfectly functional experiment system making it possible to do experiments in classroom and dormitory which guarantees usability and simplicity for students to better practice. It can be used for several courses like Network Engineering, Network Attack and Defense and Program Developing besides Computer Network. In [8], authors outline an effort at the University of Huddersfield to deploy a private IaaS cloud to enhance the student learning experience. Their paper covers the deployment methods and configurations for OpenStack along with the security provisions that were taken to deliver computer hardware. The rationales behind the provisions of virtual hardware and OS configurations have been defined in great detail supported by examples. They covers how the resource has been used within the taught courses as a Virtual Laboratory, and in the research projects. A second use case of the cloud for Automated Formative Assesment (AFA) by using JClouds and Chef for Continuous Integration is also presented.

In [9], Alves and Prata presented CSCLab, a platform implemented to run in OpenStack to manage computer science laboratories by using cloud resources which take advantage of the elastic cloud model. Using the cloud, computing resources can be allocated or released according to a greater or lesser demand. Their platform is intended to be used by non-experts in cloud technology. They described the system requirements, the architecture and the data model which uses some entities from the databases existed in the cloud services. In [10], authors proposed the use of cloud-based virtual computer laboratory which aims to release the burden of the lab's operation management and helps in the rapid construction of virtual lab. The physical servers are virtualized as a pool with virtual computers by employing OpenStack, and the management services to the class are provided for teachers and students. To further facilitate the construction of such a virtual lab, a configuration scheme of the physical resources for the virtual computer pool is presented.

Sheng et al. [11] designed a virtual experiment platform based on OpenStack which enables users to select virtual experiment resources like PC, router and servers as well as building network topology and accessing these devices. Their platform renders teachers and students a perfectly functional experiment system making it possible to do experiments in classroom and dormitory which guarantees usability and simplicity to students for better practice. In [12], authors presented a proposal for creating virtualized labs over OpenStack. Their proposal can be deployed over a University intranet, and making use of low cost PCs for end terminals, allowing for multiple uses of a PC equipped room with a high speed internet connection. They test the implementation of the presented proposal at the Department of electronics engineering of TEI of Piraeus based on two courses one Computer Networks and other Object Oriented Programming. Enhancements to the platform have also been identified, in order to focus on the students, and allowing them to store and share their lab related work and exercises.

In [13], authors described a basic overview of cloud computing, related networking technologies, several IaaS service scenarios and finally deployed private cloud in Department of InfoCom networks on University of Žilina. XU et al. [14] presented V-Lab, a cloud-based virtual laboratory education platform that provides a contained and private experiment environment for each student using the private cloud platform and SDN approaches. V-Lab also provides an interactive Web GUI for virtual resource management and a social site for knowledge sharing and contribution. The virtual resources created using OpenStack can be securely accessed through OpenVPN. The system transcends the time and space limits of traditional laboratories and provides experiments that not only allow flexible schedules but also enable students to focus on content rather than the setting up of the environment.

Our work, which shares the same target and also makes use of OpenStack, differs from these approaches in that we design and implement private cloud to fit HEI purposes. In our approach, a deployment methodology for OpenStack based private cloud using real servers is explained. Furthermore, to facilitate the construction of the private cloud, the resource requirement model is formulated by which the requirement of physical resources can be measured for a specific number of VMs.

IV. PRIVATE CLOUD DESIGN FOR HEI

We propose the following design, Fig. 2, containing a list of actions and the interactions between users' roles and OpenStack system to provide a higher education private cloud. We categorized roles specific to the stakeholders of HEI, faculty, students, researchers and administrator.

- Users Privileges: Administrator has full control over the private cloud and privileges to create and manage projects, users, roles, groups, domains, images, flavors, networks and quota assignment. Faculty can manage the course project and students in its domain. Whereas a student is assigned the member role, such as, spinning up instances, creating volumes, and creating networks based on his project. A researcher is same as student but under the control of administrator and assigned a large quota of resources.
- Course Project: Projects are organizational units in the OpenStack to which administrator can assign users. Users can be members of one or more projects. Each user-project pairing must have a role associated with it. The administrator creates and manages course projects and assigns users to them.

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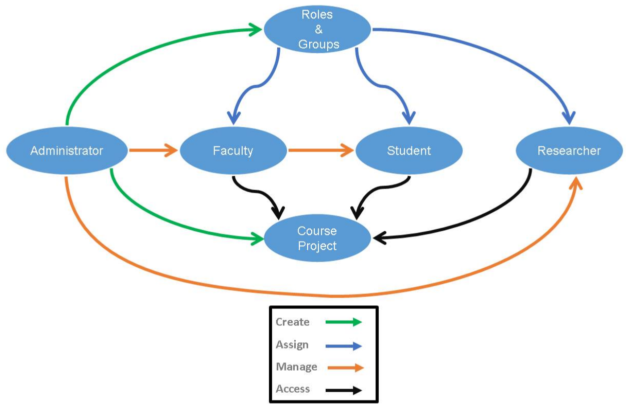

Fig. 2. HEI Private Cloud Design

- Role Management: Roles define the actions that users can perform. OpenStack uses a role-based access control (RBAC) mechanism to manage access to its resources. By default, there are two predefined roles: a member role that gets attached to a project, and an admin role to enable users to administer the environment. Three roles are created and assigned to faculty, students and researchers.
- Group Management: Groups represent a collection of users and are owned by a specific domain. Roles explicitly associate groups with projects or domains. A group role grant onto a project or domain is the equivalent of granting each individual member of the group the role of that project or domain. The addition or removal of a user to such a group will result in the automatic granting or revoking of that role to the user. Three groups are created faculty, students, and researchers, and users can be added to them.
- Quota Management: Quotas are operational limits that can be set by cloud administrator per project to optimize resources. Quotas are defined to limit the amount of vCPUs, instances, vRAM and floating IPs available to a project. This enables multiple users to use a single cloud without interfering with each other's permissions and resources. The administrator can set and manage quotas for a course project to prevent project resources from being exhausted.
- Domain Management: Domains control the administrative boundaries of Identity service entities, and aid multi-tenancy. A domain represents a grouping of users, groups, and project. A domain can have more than one project. A separate domain for faculty is created and managed by administrator.

V. DEPLOYMENT METHODOLOGY

We proposed to implement private cloud prototype with the minimal setup of two physical servers, one acting as a controller and the other as a computing node. The following discussions explain the implementation procedure followed to deploy the proof-of-concept private cloud. Summary of selected installation parameters is given in Table III.

A. Installation Parameters

1) Hardware considerations: We deploy our OpenStack based private cloud on two commodity servers. Table I contain configurations of hardware used.

2) OpenStack Distributions and Installation Tool: There are many ways to install and deploy OpenStack. There are several commercial and non-commercial software distributions powered by OpenStack. Non-commercial linux distributions includes Ubuntu, CentOS and OpenSuse operating systems. We have selected *RPM Distribution of OpenStack* (RDO) for our deployment. RDO can be deployed on top of CentOS. RDO is a freely-available, community-supported Red Hat's distribution of OpenStack. The installation tool Packstack specific to the RDO is used. Packstack is a command-line based installation tool that enables rapid deployment of OpenStack on existing servers [15].

3) Hypervisor selection: KVM, Xen, Hyper-V, and VMware are the hypervisors available for use with Nova. The KVM hypervisor is the most broadly used hypervisor on Linux and we selected same for our deployment.

4) Service Selection: The OpenStack system consists of several key services as discussed in section 2. The administrator can install and configure required services separately or together. These services are required depending on cloud needs. We opted for Dashboard, Compute, Identity, Networking, Image, Block Storage and Object Storage services.

5) Network Design: In our network design, Fig. 3., the controller node is used to run the dashboard, identity service, networking and the image service. The compute node is used to run compute service, block storage and object storage service. The provider network is used to interact with the host using SSH as well as the OpenStack API and the dashboard. Provider network gives users the ability to access their instances directly by their fixed IP addresses. Management network is used by OpenStack neutron service for management level interaction between physical servers and virtual machines. The public network is used to access the OpenStack services using the web interface. An instance assign floating IP can also connect to internet and access by the user from remote locations.

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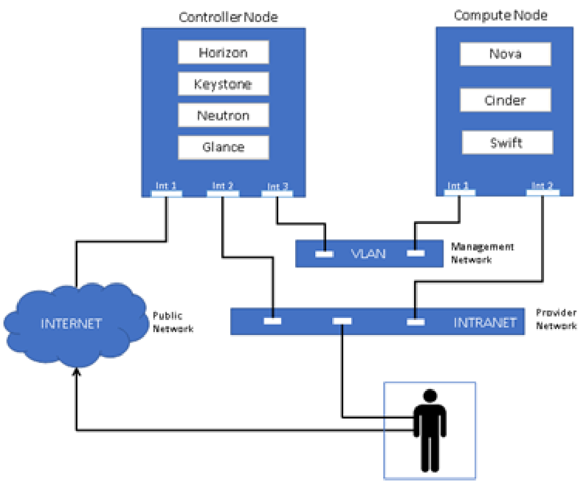

Fig. 3. Network Design

B. Resource Requirement Model

Calculating the capacity of cloud involve defining the standard instance sizes and deciding the overcommit ratio. Instance sizes are referred as flavors in the OpenStack. The over-commit ratio is the proportion of available virtual resources compared to the existing physical resources. Physical servers have their own virtualization capabilities and declared by manufactures in the product documentation.

1) Flavor Creation: In OpenStack flavors describe the hardware configurations. A flavor has these prominent parameters, the number of vCPUs, the amount of memory vRAM, the amount of root disk storage and ephemeral disk. When a user deploys instance they can select a flavor for hardware configuration. We have defined five flavors based on our course workloads. Listed in Table II.

2) Overcommit Ratio: OpenStack allows overcommitting CPU and RAM on compute nodes to increase the number of instances running on a cloud. As per our hardware resources, we used CPU allocation ratio: 10:1 and RAM allocation ratio: 1.5:1. Thus, for our deployment, a compute node with 8 physical cores can support up to 80 virtual cores of compute and 32 GB of RAM can support 48 GB vRAM.

To calculate how much infrastructure needed to support a private cloud, we deduce the following equations.

- Equation (1) is used to calculate the expected number of instances possible on a compute node.

$$\frac{\text{CPU Overcommit ratio} \times \text{Physical cores} \times \text{Numbers of CPU}}{\text{Virtual core}} \quad (1)$$

- Equation (2) is used to calculate expected size of RAM required for possible numbers of instances.

$$\frac{\text{Numbers of instaces} \times \text{instance RAM size}}{\text{RAM Overcommit ratio}} \quad (2)$$

For a case, f.medium flavor uses 2 vCPU, and 2GB of vRAM. A server with 1 CPU of 8 cores, with hyperthreading turned on, the default CPU overcommit ratio of 10:1 allows for ((10 × 8 × 1) / 2) i.e., total 40 f.medium flavor instances. The default memory over-commit ratio of 1.5:1 means that the server needs at least (40 × 2048 MB / 1.5), i.e., 54 GB of RAM.

Since, our compute node contain 8 core single processor and 32 GB of RAM, it can support up to 48 GB of vRAM and 80 vCPU. We decide to have a quota of 20 f.small, 10 f.normal, 5 f.medium, 3 f.average and 2 f.large number of instances for our private cloud. It will consume 48 GB of vRAM and 54 vCPU.

VI. CONCLUSION

Cloud computing is changing the world of IT. It has completely transformed the way data center works. Higher education institutions are also looking into it to take benefits of its potential. OpenStack provides IaaS for hardware resources and standard mechanism to manage and distribute virtual machines. We designed and implemented a private cloud for HEI using OpenStack which enables students and researchers to work with ready to use pre-configured virtual machines as per course requirement and create their own virtual resources like servers, routers, and network topology. Our platform delivers students and researchers a functional system to do experiments in a sandbox environment. In future, we are looking into leveraging the OpenStack private cloud for providing platform to learn SDN and NFV technologies.

[redacted]

Table I. Hardware Configurations

S. No.	Hostname	Model	CPU cores	Memory	Disk	Network
1	Controller Node	Hp Proliant Server Dl-380-G9 - Rack	8Core/16 Threads	16GB (1*16GB)	3x300 GB	HP 1Gb 331i Ethernet Adapter 4 Ports
2	Compute Node	Hp Proliant Server Dl-380-G9 - Rack	8Core/16 Threads	32GB (2*16GB)	3x300 GB	HP 1Gb 331i Ethernet Adapter 4 Ports

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Table II. Flavors Configurations

S. No.	Name	vCPU	vRAM	Root Disk	Ephemeral Disk
1	f.small	1	500 MB	5 GB	0 GB
2	f.normal	1	1 GB	10 GB	20 GB
3	f.medium	2	2 GB	20 GB	30 GB
4	f.average	2	3 GB	30 GB	40 GB
5	f.large	4	4 GB	40 GB	60 GB

Table III. Selected Installation Parameters

Installation Requirements	Selected Parameters
Base Operating System	CentOS 7
OpenStack Distributions	RedHats' RDO (RPM Distribution of OpenStack)
Installation Tool	PackStack
Hypervisor	KVM
No. of Nodes	2 (one controller node and one compute node)
OpenStack Version	Pike
OpenStack Services	Dashboard, Compute, Identity, Networking, Image, Block Storage, Object Storage
Flavor Size	f.small (1 X 0.5), f.normal (1 X 1), f.medium (2 X 2), f.average (2 X 3), f.large (4 X 4)
Overcommit Ratio	CPU 10:1, RAM 1.5:1
Network Design	Flat

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